THE UNIVERSITY OF BRITISH COLUMBIA
CERTIFICATE PROGRAM IN REAL PROPERTY ASSESSMENT /
DIPLOMA PROGRAM IN URBAN LAND ECONOMICS
BUSI 444 CASE STUDY EXAMINATION
January 5, 2021

GENERAL INFORMATION

This Case Study Examination requires you to create valuation models based on a database of property sales. You must also write an appraisal report to describe your results and the procedures used to obtain these results. This report is heavily based on two of the Projects completed during the term of BUSI 444.

The appraisal report for the examination should provide the Assessor of Marilu City with a mass appraisal model which could be applied to value the properties contained in the database provided to you. The valuation date is July 1, 2017 and the 2018 Assessment Roll is to be presented to Marilu City on December 31, 2017. Assume that your report will be presented to the Assessor on November 30, 2017. The Assessor will have one month from that date to take any action that you recommend, to not only apply the model you have developed, but also carry out any further actions you recommend as necessary to improve the resulting assessments.

Assume that the Marilu City is on a full market value assessment base. Therefore, the model that you recommend to the Assessor must reflect the full actual (market) value of the fee simple interest in the properties.

Submissions for this examination should follow the standard format for the appraisal process and appraisal reports. This format is described later in this document, see the section titled "Mass Appraisal Report Outline".

As this is an examination, you must describe the processes you are using in order to demonstrate to the reader of the report that you understand mass appraisal theory and know how to develop a sound, practical valuation model. However, as much as possible, the examination should be presented in the format of an actual report. The only statistics that should be included in the body of the report are those necessary to prove the validity of the opinion being expressed. The full detail reports should only be included as appendices to the main report. For example, if the mean and standard deviation obtained from the descriptive statistics report are included in the body of the report, these should be followed by a reference to the full table of results (i.e., "Full details of this report may be found in the descriptive statistics table contained on page 3 of Appendix A").

Note that the syntax files showing all transformations MUST also be included in your appendix. Syntax files should contain ONLY the transformations not the commands to generate reports, charts, or other output.

BACKGROUND INFORMATION FOR MARILU CITY

General Information

Marilu is a mid-sized city located in Northern Ontario. The 2016 census population was 70,563. Due to its northern location Marilu has long cold winters (mean January temperature -12° C) and short summers (mean July temperature 15° C). Average annual precipitation is 63 cm with snowfall of 240 cm.

Forestry is the major industry in the area. The region has a large number of sawmills and several pulp mills. The town is served by two railways and has a class A airport. The town is a major centre for government, health, and education services with a regional college and regional hospital. It is the commercial centre for a large outlying region with several regional shopping malls and 180,000 square metres of commercial floor space.

Real Estate Information

The residential real estate market contains approximately 24,000 dwelling units. Of these dwellings 70% are owned and 30% are rented. Single detached homes make up about 65% of the existing housing stock, while apartments over five stories account for only two percent and mobile homes about five percent. The market has been quite buoyant in recent years and is expected to continue that trend for the next few years.

One-storey homes make up about 80% of MLS sales with 1½ and two-storey homes making up about five percent and condominiums and mobile homes the remainder. The sales data provided is taken from the densely developed urban core of the city.

Neighbourhood Descriptions

Neighbourhood 713 is located directly to the west of the commercial core of the city. Most of the neighbourhood is laid out in a grid pattern with services located in rear alleys. Housing stock in this neighbourhood is among the oldest in the city and there is some commercial encroachment in the eastern section of the neighbourhood. There are more sidewalks in this neighbourhood than others and the housing is generally made up of 80 and 90 classes. Due to the alleys many homes have detached garages with access from the alley rather than from the street. This can be a problem in the winter as often the alleys are not kept clear of snow.

Neighbourhood 715 is located to the west of neighbourhood 713 and has two sections of diverse housing stock. To the north of 5th Avenue the homes are generally older on streets laid out in a grid pattern; some homes on the northern fringe have views of the river. The southern section has more newer homes than the northern section, generally built in the 1970s in a more modern layout on curved streets with cul-de-sacs with underground services. The southern fringe along 15th Avenue, a major bus route, is developed mainly with apartment blocks.

Neighbourhood 716 is north and west of neighbourhood 715. It was developed at about the same time with a modern layout and underground services, however, a few of the streets have rear alleys. The northern streets have some properties with river views. Housing quality in this neighbourhood is generally better than most neighbourhoods. The majority of the sales of class 145 and better are in this neighbourhood.

Neighbourhood 731 is located to the east of the central business district. It was developed mainly in the 1950s with many good quality homes along the river, but with the development of the pulp mills up river in the 1970s these homes lost their appeal due the offensive odour from the mills. The layout of the neighbourhood is a grid pattern with alleys and overhead wiring. There are two major parks in the neighbourhood.

Neighbourhood 732 is a small neighbourhood located south of 15th Avenue and north of the major park containing most of the baseball and soccer fields in the urban area. This is developed to the modern standard layout but for the most part has overhead wiring. It is very popular with doctors and teachers as it is very close to the regional hospital, the regional college, and the major high school in the city. Due to this there ae usually very few sales in this neighbourhood. The housing stock is mainly one-storey modern standard 90 and 140 class houses.

Neighbourhood 734 is another mixed quality area. The western portion is low lying and subject to the adverse effect of the pulp mill emissions. The streets are mainly grid pattern with rear alleys and overhead wires. Housing stock in this area is low quality with many basement suites. The eastern section is on higher ground and does not suffer as greatly from the pulp mill emissions, streets here are modern standard with underground services and housing quality is generally better here than in the other section of this neighbourhood.

Neighbourhood 736 is in the southeastern corner of the urban area, it has some parts with underground service and some with overhead wiring. The housing stock is generally modern standard 90 and 140 class homes.

Neighbourhood 737 is in the southwestern corner of the urban area. It was primarily developed in the 1970s with most homes in the 90 and 140 classes. The general layout is modern standard with underground wiring, cul-de-sacs, and few through streets. However, there are a few streets that have rear alleys similar to neighbourhood 716.

EXAMINATION REQUIREMENTS

You are to develop two models for possible use by the Assessor of Marilu City. One model will be based on the cost approach to value and the other will be a direct sales comparison-based approach.

1. Describe, and state using standard mathematical symbols, an appropriate general model to use as a market adjusted cost approach.

2. Review the available variables in the database, and the cost data provided in the Course Workbook, to select the variables suitable for this model type. This may require you to transform some variables.

3. Specify the cost model.

4. Develop a market-based land value using the land residual technique.

5. Calibrate the cost model using the land value from Step 4 above, the cost data provided in the Course Workbook, and the depreciation information obtained from the development of the PCTGOOD variable.

6. Develop and apply any necessary market adjustments.

7. Describe, and state using standard mathematical symbols, an appropriate general model to use as an additive multiple linear regression model for the direct sales comparison approach.

8. Review the available variables in the database using all analytic techniques covered in the course to select the variables suitable for this model type. This may require you to transform some variables.

9. Specify the direct sales comparison model.

10. Calibrate the direct sales comparison model using an additive multiple linear regression.

11. Completely test and evaluate both models for use as described in the General Information section the "Mass Appraisal Report Outline" presented later in this document. The quality of the models must be evaluated from both an appraisal and an assessment perspective.

12. State your conclusions as to model quality and provide recommendations to the Assessor of Marilu City as to which model will most effectively and accurately provide the necessary values. In addition, make recommendations for any necessary actions the Assessor should take before using the model to value these neighbourhoods for the 2018 Assessment Roll.

Helpful Hints:

- You should use, as the basis for the exam submission, the work already done on Project 1 and Project 2 for BUSI 444. It is expected that the marking feedback on the graded Projects will be incorporated into the exam submission.

- The main **new** work beyond the projects will be to create a solid report that links the two projects, fully tests both the cost and direct sales comparison models, makes adjustments as needed, outlines recommendations, and provides conclusions for the report.

- You should aim to have your model, at least, meet IAAO's minimum requirements of CODs less than 15% and median ASRs between 90% and 110%.

GRADING INFORMATION

Since this is an examination in the form of a demonstration appraisal report, you are expected to submit a report that follows as closely as possible a standard single property appraisal report both in terms of format and content. In this report, you must demonstrate your knowledge of appraisal theory and sound statistical practice in developing and testing your mass appraisal model.

In grading these exams, the reports are divided into sections with mark allocations as shown below:

1.	Appraisal Foundations and Report Quality	20
2.	Cost Model	20
3.	Transformations and Data Analysis	15
4.	Variable Selection	15
5.	Direct Sales Comparison Model Calibration	10
6.	Testing of the Models and Conclusions	20

The final analysis and conclusion should include a complete analysis of both models and comparison of these results with a recommendation and selection of the better model for use.

The marker may also identify a "major fault" which will result in a failing grade for the Case Study Examination regardless of the grade indicated from the sections above. A major fault would typically be an error that is of a degree of seriousness that the entire modelling exercise is undermined as a demonstration of competency. Examples of a major fault would be poor coverage of the appraisal foundations OR massive multicollinearity among variables in the variable selection process for the regression-based model. Another major fault is using the wrong database in completing your examination.

You are required to achieve an overall course grade of 60% for the purpose of successfully completing the course. In order to receive professional recognition towards a demonstration report requirement, a grade of 70% must be attained on the final Case Study Examination. In order to receive credit from the Appraisal Institute of Canada, a minimum of 50% must also be achieved on each section outlined above.

ATTENTION STUDENTS: INDIVIDUAL WORK REQUIREMENTS FOR CASE STUDY

This Case Study submission is an **examination** that will count for credit towards completion of a certificate or diploma program and may also qualify you for accreditation by professional valuation associations. Because of these important consequences, it is vital that you produce and submit work which is entirely your own. Studying in groups is not allowed for this examination – answers must be produced on an individual basis. The marker of the Case Study Examination must be able to certify that you have a reasonable understanding of the concepts and techniques employed in mass appraisal. For example, you must:

- describe your own views on the appraisal problem, how the background information will affect the appraisal model's creation, and the methodology you will apply in creating this model;
- create your own mass appraisal models, applying your personal appraisal assumptions and decisions;
- submit your own individually-produced document(s) to show how the models were created; and
- provide your own summary of findings and discussion of the validity of your models.

(**Note:** this is only a partial list of what must be submitted for a complete Case Study Examination. For a more comprehensive list, see the section below titled "Mass Appraisal Report Outline")

It is unacceptable to copy someone else's work and submit it as your own. This would include copying solutions from old answer guides or copying from other students' examinations. Any recognized case of copying in the Case Study Examination is considered plagiarism and could result in one or more of the following consequences for you:

(a) an official reprimand letter sent to you from the Real Estate Division, a copy of which will be kept in your file at the Real Estate Division;
(b) having to re-do and resubmit the examination in question or complete a replacement examination provided by the Real Estate Division;
(c) losing the right to any credit for the examination toward completion of the course or toward professional accreditation; **OR**
(d) expulsion/suspension from the program.

You are required to fill out and sign the Declaration on the following page and include it with your examination submission. Since the exam is to be submitted electronically, two suggestions for doing this are: one – attach an electronic signature to the document; or, two – print the Declaration, sign it, scan the signed document, and include it as the final page of your submission.

DATABASE DESCRIPTION

The database for the examination can be found on the BUSI 444 Online Readings webpage under 'CASE STUDY EXAM'**.** Please note that the database provided is a fresh copy of the Marilu database that was used in Project 1.

You may reference the BUSI 444 Course Workbook, pages **Projects.5** to **Projects.10**, for a complete description of the exam (Marilu) database.

> PLEASE NOTE: there is a sufficient number of sales in the database provided to allow the creation of a MODEL and TEST database for the direct sales comparison approach to value. You should develop the direct sales comparison approach to value using a MODEL database of at least 300 sales and test your results on a TEST database of at least 60 sales and no more than 150 sales. There is a RANDOM variable included in the database to enable you to select a random sample of sales for the MODEL and TEST files. The database can be sorted in descending sequence by the RANDOM variable – ensure that the database is sorted by this variable before creating the MODEL and TEST databases.
>
> When testing the model, you should use the TEST database ONLY to ensure that the mean and median are acceptable, and the dispersion, as measured by the COD, is satisfactory. The model should then be applied to the FULL exam database for complete testing. This latter step is required in order to ensure that there are sufficient sales available for all stratifications, so that results will not be biased by the small sample size in the TEST database. No adjustments should be made based on the testing using the TEST database.

COST DATA

You are to reference the BUSI 444 Course Workbook, pages **Projects.10** to **Projects.11**, for the cost data required for the exam.

DECLARATION

THIS DECLARATION MUST BE SIGNED AND INCLUDED WITH YOUR EXAMINATION SUBMISSION

Name: ____

Student Number: ______________________

I warrant that I am the sole author of this Case Study Examination; I have written all narrative sections and created all aspects of the model on my own. I have not knowingly plagiarized or carried out any other form of academic misconduct and have submitted this examination in good faith.

____
(Signature)

____
(Date)

MASS APPRAISAL REPORT OUTLINE

The final examination for BUSI 444 requires you to create a mass appraisal model given property cost and sales data and to prepare a report outlining this process and describing the results obtained. The report must provide the level of detail that an Assessor would need to apply this valuation model in a real assessment context. Therefore, the report must contain all pertinent information and must be organized in a standard appraisal report format. The analysis for the exam is heavily based on the work you have already done during Project 1 and Project 2 for BUSI 444.

As an appropriate published guide for mass appraisal reports is not readily available, the following outline has been provided to describe the report format and content expected.

GENERAL INFORMATION

The main difference between single property appraisal reports and mass appraisal reports in the introductory sections is the scale of the mass appraisal report. Where a single property report will contain a neighbourhood analysis and a description of the subject property, a mass appraisal report will describe all of the neighbourhoods that are covered by the report and, in a general way, all of the properties to be appraised.

For example, a mass appraisal report requires a description of the properties to be appraised; this should be something similar to this excerpt from a report based on a rural area:

> The properties to be appraised are all the rural residential small holdings, less than five acres, in neighbourhoods 200, 230, 250, 320, and 470. These properties vary in age from five to over 100 years, and the level of maintenance and physical condition is also mixed. Neighbourhood 200 and 230 are the closest to the city and are undergoing substantial "gentrification", with most of these small holdings owned by professional people employed in the city. There are a large number of very large high-quality homes with substantial outbuildings and riding rings for show horses and other equestrian activities. There is continuing demand for this lifestyle and not much space left in these neighbourhoods for further development. This has led to increasing land values and may start to put pressure on the adjoining neighbourhoods. The remaining three neighbourhoods are located farther from the city, and the primary motive for living in these neighbourhoods would seem to be the lack of regulation in terms of construction quality and types of dwelling units, together with a relatively low tax rate. The homes in these neighbourhoods are a mix of styles and suffer the most from deferred maintenance and lack of finish. Many of the homes were originally manufactured homes which now have a patchwork of additions in varying stages of completion. There are also many old farm buildings that remain in their original locations after the subdivision of the farm into smaller holdings. Many of these have been well-maintained, and generally represent the better-quality homes in these three neighbourhoods.

In single property appraisal reports, the appraiser normally describes the purpose of the appraisal (i.e., mortgage, expropriation), the definition of value to be used (fee simple, leasehold, value to owner for expropriation) and the valuation date. Mass appraisal reports also require these features. The definition of value is particularly important as this varies from place to place and assessed value and market value for assessment purposes must be clearly defined in the report (if market value has any bearing on the assessment in this jurisdiction).

The following is an excerpt from a mass appraisal report which describes the purpose of the appraisal and the definition of value to be used:

> This report contains estimates of property values for the Village of Moose Elbow, based on the statutory requirements for property tax assessment in the Province of Artica. The value of land and buildings are stated separately on the assessment roll. Land value is based on highest and best use of the land at the date of assessment, except for farm land which is valued in agricultural use only. The

market value of the land (as vacant) is based on market evidence obtained from property sales occurring in the calendar year prior to the year of assessment. The value of the buildings is based on the assessment cost manual published by the Province of Artica Assessment Standards branch, adjusted for local market conditions. However, the total value of the land and buildings cannot exceed the fair market value that comparable properties would sell for in the calendar year preceding the valuation date, when considering the highest and best use of the property on the valuation date.

REPORT OUTLINE

The appraisal report should contain the following, either as separate sections or items, or as headings and sub-headings within a section. Please see the Grading Information on Page 4 above.

Letter of Transmittal & Table of Contents	
Letter of Transmittal	• This is a letter to the person who initiated the preparation of the appraisal report. It should summarize the appraiser's understanding of the scope of the work that was to be completed and summarize the general conclusions of the report. It should be the first page of the report. **The letter should be signed.**
Table of Contents	• Mass appraisal reports are normally quite long. A table of contents provides the reader with a reference to where specific material is located in the report so that portions may be easily reviewed. It is also courteous, in a long report, to provide tabs on the main sections so that they can be easily found.
Main body of the report Sections 1 Through 5. For the purpose of a demonstration appraisal report, you must expand on each of these topics to show that you fully understand both the theory and practical aspects of mass appraisal. Any figures or tables that are included in the main body of the report should be numbered; items appearing the Appendix should be clearly referenced.	
Section 1 - Definition of the Appraisal Problem	
Properties to be appraised	• The properties being valued, are a key component for any appraisal assignment in order to define the scope of work required. • In a single property appraisal this would be the subject property. • In a mass appraisal report it is the group of properties.
Property rights involved	• The definition of value is particularly important and this varies across jurisdictions. • The assessed value and/or market value for assessment purposes as defined within the jurisdiction's legislation must be clearly defined in the report. • The property rights on which the values are based (i.e., fee simple, leasehold, value to owner for expropriation) needs to be identified and defined.
Purpose of the appraisal	• In single property appraisal reports, the appraiser normally describes the purpose of the appraisal (e.g., mortgage, expropriation). • For this report the intended user of the report is the Assessor for Marilu, therefore the purpose and function (how the report will be used) needs to reflect this.
Date of appraisal (valuation date)	• The reference date and/or the valuation date, provides the time reference to the appraisal report's analysis and valuations.
Preliminary planning for approaches & data collection	• Data collection and sales editing techniques are typically a requirement for a mass appraisal report, or in a demonstration report. You would be expected to find and edit your own data, the actual data collection and editing techniques would have to be described in detail. • For this demonstration report a complete edited database is provided, so data collection and editing can only be discussed in a theoretical or hypothetical way. You should state the name of the database used and briefly outline its characteristics (e.g., number of observations, number of variables).

Section 2 - Preliminary Survey and Analysis	
Approaches to Value	• Discussion of all three valuation approaches should be included here with reasons for using some and rejecting others for this report.
Data Requirements	• Data requirements of the selected approaches. • In this section, there should be a discussion of how the data requirements differ from single property appraisal, particularly with respect to adjustments in the direct sales comparison approach.
Section 3 - Data Collection, Market Analysis, and Highest and Best Use Analysis	
Data Collection and Market Analysis	• Market analysis should discuss the four forces that influence value: economic, social, environmental, and governmental factors, which affect the supply and demand for real estate within the market area. • Characteristics of neighbourhoods should be examined, with an analysis of possible commercial encroachment and other factors which could affect the highest and best use of these properties. • Use statistical analysis to report housing characteristics such as mean and median size, age, and sale price by neighbourhood or other relevant property groupings. • The primary emphasis of the analysis is to tie the four forces back to the properties being appraised for the highest and best use conclusion.
Highest and Best Use Analysis	• It should be clear to the reader how the Highest and Best Use needs in mass appraisal are similar to or differ from that required in single property appraisal. • The Highest and Best Use Analysis needs to tie back to and be based on the analysis of the data collection and market analysis. • Highest and Best Use conclusion determines how the properties will be valued.
Section 4 - Valuation using Model Building and Model Calibration (cost model and sales comparison model)	
Data Analysis and Transformations (for each model)	• A review of the data provided needs to completed. A time adjustment should be carried out. Any corrective transformations required to include the variables into the model need to be fully explained. • Include your decisions, rationale, and/or justification within the main body and the detailed output within the Appendix. • A more detailed examination of the data will be required for the direct sales comparison model over the cost model.
Model Specification (for each model)	• Mass appraisal reports must completely describe any valuation models used. In addition, the report should explain why any variables were included or excluded in the specification process, particularly if the variables would normally be considered factors in the purchase and sale decisions of buyers and sellers. • Include a detailed analysis of the model specification to be used, including reasons for selecting one model structure over other available options. • It is NOT necessary to include detailed variable selection reports in the main body of the appraisal report. Only the variable selection reports which illustrate differences in techniques attempted should be included, and these should only be found in an appendix to the main report. • Discussions or explanations similar to that above should be provided for any other variables where the form of the variable can be either additive or multiplicative or where variables can be entered as binary or area variables or any other cases where choice of variables must be justified. • Variables should be tested for multicollinearity (direct sales comparison model only). Independent variables selected for the model should not demonstrate a high level of correlation with each other. • Following this justification of the variables used, the specified model should be given in mathematical form.

Model Calibration (for each model)	• In describing model calibration, there should be explanation of the coefficients from an appraisal standpoint (e.g., relative size of the coefficient and the relationship to the expected "market value" of that variable). • When calibrating the direct sales comparison model, the detailed correlation reports, residual reports, and so forth, should again be included in an appendix. • The main body of the appraisal report should only include the final regression reports (Model Summary, ANOVA, and Coefficients table). • The main statistics contained in these reports should be briefly explained.
Section 5 – Final Analysis (including full testing of both models)	
Model Testing	• When only one model is developed, there is no need to reconcile the valuation. However, even if only one model is developed, it is necessary to comment on the statistical criterion to select the best model. The important statistics for the direct sales comparison model are R2, Adjusted R2, standard error of estimate (SEE), F-statistic, VIF values, and t-statistics. • The details of any of these statistical tests should be included in an appendix to the main report, so that they can be carefully examined by assessment staff. For the general reader, the main report can include summaries of these tests in table form.
Quality Control	• All models should be subject to complete testing for appraisal level and dispersion, and these results should be compared to local and international (IAAO) standards. • In addition, complete tests for horizontal and vertical equity (Ratio Statistics) must be conducted to ensure that the level of assessment is consistent for all types of property characteristics, age groups, price groups, neighbourhoods, and property types, if applicable. • Again, summary tables of results should be included in the main report and detailed statistical reports should be placed in an appendix. • Any necessary adjustments to the model should be made.
Conclusions	The final section of the report should include: • Conclusions concerning the quality of the model or models developed. The quality should be discussed from both an appraisal and an assessment perspective. • A discussion of any potential problems that may arise in the use of the model(s). • An outline of any data problems or inconsistencies found during the analysis or testing stages. • If required, provide any recommended corrective action that can be taken during re-inspections. • Recommendations for model use. • These conclusions should also be summarized in the letter of transmittal, which should be the first page of the report.
Appendix	
• All detailed reports need to be shown in the Appendix and should be well-referenced within the main body of the report using headings and page numbers. Anything appearing in the Appendix should be referenced in the main report. • Syntax: the syntax statements for all transformations must be included. Note: Include your conclusions and report decisions/summaries in the main body of the report and not in the Appendix.	

Overall Report Presentation and Style
The submission will be graded with consideration to: • Composition, coherence, consistency, and style • Spelling and grammar • Organization
Remember; the objective of this mass appraisal report is to lead the Assessor from Marilu City through your process in developing the valuation model to be applied to the jurisdiction. It should be obvious to the Assessor the steps, analysis, rationale and supporting documentation (in the Appendix) you used to develop and justify your mass appraisal report.

LENGTH OF REPORT

A complete report will contain approximately 40 pages of report detail and will be supported by an appendix which contains all SPSS reports that were relied on to justify any decisions or reach any conclusions, including the syntax statements showing all transformations. The Appendix should be organized and indexed and would contain, on average, approximately 150 pages. Note that these are only rough limits and that you should provide all **relevant** information, i.e., include anything that you used to substantiate decisions or adjustments. Note also that a very well written and organized report could require significantly less than these numbers of pages.

The table below summarizes the suggested size of submissions and the maximum allowable pages. Under no circumstances should you need more than the stated maximum number of pages for your report. If your submission exceeds these limits, EITHER: 1) you will receive an email stating that you need to reduce the size of your submission; OR 2) the additional pages will not be read by the marker and marks will be deducted for their inclusion.

	Report	**Appendices**
Suggested	40 pages	150 pages
Maximum	60 pages	200 pages

Please keep in mind that the intent of this Case Study submission is to simulate the type of report that you would actually deliver to the Assessor of Marilu. You are acting as an appraiser in an assessment office and your report will be read by someone who has some general appraisal knowledge, but who is likely not familiar with statistics or mass appraisal theory. You must write your report specifically to address the information needs of this person so that they can understand and apply your work. Submitting an excessively long report to your assessor supervisor, e.g. a 100-page report along with 300 pages of SPSS output, is likely to be similarly inappropriate to a report which is much too short.

Finally, you are also reminded that, in addition to a practical demonstration of mass appraisal processes, this is an examination of the underlying theory. You must ensure that you provide sufficient explanation of the process and procedures in order to demonstrate your knowledge of sound mass appraisal theory.

SUBMITTING THE CASE STUDY EXAMINATION

Below are some specific formatting guidelines for preparing your Case Study Examination:

- font of 11-point or larger, 12-point preferred; except where a smaller font is required for some SPSS reports to fit on a page;
- minimum 1-inch margins;
- pages numbered consecutively in the top right corner, separately numbered for the report body and appendix;

- all sources of information relied upon in producing your submission should be carefully documented in a bibliography or list of references.

Ensure that you include the syntax statements for all of your transformations, so the marker can review the transformations you carried out in preparing your report.

Do not forget to include **and sign** the Letter of Transmittal and the Declaration.

Note that your Case Study Examination must be submitted via Turnitin. There are three Case Study related assignments specified within Turnitin:

1. Case Study Exam – Report: use this if your report is all in one document.
2. Case Study Exam – Appendix (if needed): use this if you have your appendix saved as a separate document.
3. Case Study Exam – Syntax (if needed): use this if you have your Syntax saved as a separate document.

Turnitin will accept a document that is up to 40 MB in size. If your document is larger than this, you may wish to try saving it as a PDF, as this may reduce the size considerably. If you experience any difficulty in uploading your Case Study document(s) please contact your tutor.

Please ensure that any documents submitted to Turnitin are clearly titled with your last name and student number so that they can be identified as belonging to you.

Your Case Study Exam should be graded within six weeks of its submission. Your grading results can be viewed in Turnitin. Out of Turnitin you can (and should) download the pdf grading document for your records.